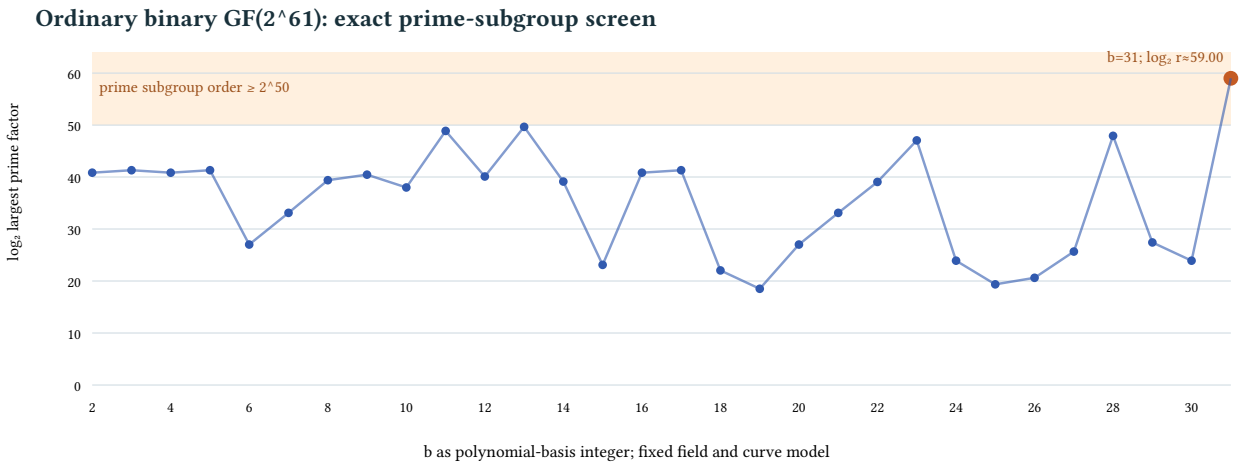
Ordinary binary subgroup input near 2^59

Input-selection screen, 2026-10-17 UTC. Fix GF(2^61) with polynomial t^61+t^5+t²+t+1 and E_b: y²+xy=x³+b. Inspect b=2..33 in order; stop at the first curve with a certified prime factor r in [2^50,2^63]. The screen stopped at b=31 after 30 attempts. Every attempted group order was fully factored, each factor checked prime, and the product reconstructed.



For b=31=z⁴+z³+z²+z+1, j=1/b lies outside F_2. This ordinary binary curve is not descended from F_2. Its group order is 2,305,843,010,228,847,052 = 4 × 576,460,752,557,211,763, where r=576,460,752,557,211,763 is prime ($\log_2 r \approx 59.00$).

The polynomial-basis source point P=(1,446358225178544464) projects by cofactor four to G=(938161616493806685,1665004790792120888). Independent Sage arithmetic checked irreducibility, primality, 4P=G, rG=O and curve membership. Since G has order r and Hasse's interval contains only one multiple of r, namely 4r, this also certifies the full curve order.

Stage	Current evidence
Curve and subgroup	Explicit generator; independent Sage replay
Factor base, m=2…8, direct S_8	Unmeasured
Relations, rank, LA, target	Unmeasured
Complete IC and matched rho	Unmeasured

This is an input fixture, not a speed comparison. The 30-curve list is a deterministic bounded search, not a random sample. The direct F_2-x rank gate for curves descended from F_2 does not apply to this curve. A frozen target-blind base and complete experiment remain required.

Data: research/icperf_ordinary_binary_fixture_20261017/ Screen, fixture, and Sage replay are source- and input-hash bound.